

Computational Economics: Problem Set 8

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1 Economic Model

Following [1], the firm chooses its capital stock in the presence of market power and convex adjustment costs. The value function $V(K, A)$ solves

$$V(K, A) = \max_{K'} \{ \pi(K, A) - p(K' - (1 - \delta)K) - C(K', K) + \beta \mathbb{E}_{A'|A} [V(K', A')] \}. \quad (1)$$

for all (K, A) , where K is the current capital stock and A is a measure of profitability.

Profits are given as follow

$$\pi(K, A) = AK^\alpha, \quad (2)$$

where α measures the degree of market power. Capital evolves according to $K' = (1 - \delta)K + I$, and adjustment costs are quadratic in investment relative to the capital stock:

$$C(K', K) = \frac{\gamma}{2} \left(\frac{K' - (1 - \delta)K}{K} \right)^2 K. \quad (3)$$

The profitability shock A follows a stochastic process governed by parameters (ρ, σ) . [1] treat (ρ, σ) as parameters of the process for the firm-specific shocks to A , and approximates this process using a discrete state space constructed by the method of [2]. In the PS8 instructions, this is implemented as an AR(1) in logs, $\ln A_{t+1} = \rho \ln A_t + \varepsilon_{t+1}$ with $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2)$, before discretization. (Followed by PS8 instruction, which specifies that firm productivity follows an AR(1) process in logs and that you should discretise it using the Tauchen (1986) method.)

1.1 Costly External Finance

Section 3.2 of [1] introduces costly external finance. The cost of accessing capital markets is

$$\phi(E) = \phi_0 + \phi_1 E, \quad (4)$$

where E measures the extent of external finance and is defined by

$$E \equiv I - \pi(A, K). \quad (5)$$

With this cost of capital-market participation, the firm's dynamic problem becomes

$$V(K, A) = \max\{V^e(K, A), V^i(K, A)\}, \quad (6)$$

where V^e and V^i are the value functions under external and internal finance, respectively. [1] write the external-finance problem as

$$V^e(K, A) = \max_{K' > \pi(K, A) + (1-\delta)K} \left\{ \pi(K, A) - p(K' - (1-\delta)K) - C(K', K) - \phi_0 - \phi_1 p(K' - K(1-\delta)) - \pi(K, A) \right\} + \beta \mathbb{E}_{A'|A}[V(K', A')], \quad (7)$$

and the internal-finance problem as

$$V^i(K, A) = \max_{K' \leq \pi(K, A) + (1-\delta)K} \left\{ \pi(K, A) - p(K' - K(1-\delta)) - C(K', K) + \beta \mathbb{E}_{A'|A}[V(K', A')] \right\} \quad (8)$$

for all (K, A) .

For the specification in Table 3, the authors define the fixed cost relative to mean capital, $\tilde{\phi}_0$, and estimate the parameter vector

$$\Theta = (\alpha, \gamma, \rho, \sigma, \tilde{\phi}_0),$$

and set to $\phi_1 = 0$.

2 Estimation Strategy

The problem set report's goal (objective) is to replicate the costly external finance part of Table 3 using a Simulated Method of Moments (SMM) estimator. Let $\hat{\psi}^d$ denote the vector of data moments reported in Table 3(b) and $\hat{\psi}^s(\theta)$ the corresponding moments computed from simulated panels generated by the model. The SMM estimator solves

$$\hat{\theta}_{S,T}(W) = \arg \min_{\theta} \left[\hat{\psi}^d - \hat{\psi}^s(\theta) \right]^\top W \left[\hat{\psi}^d - \hat{\psi}^s(\theta) \right], \quad (9)$$

where W is a positive definite weighting matrix.

PS8 instruction specifies that first estimate the model with W equal to the identity matrix, and then construct an optimal weighting matrix from the variance-covariance matrix of the simulated moments using multiple simulations at the first-stage estimate.

3 Results

The estimation results are presented in Table 1. The estimated fixed cost of external finance, $\tilde{\phi}_0$, is statistically indistinguishable from zero from this simulation experiment.

Table 1: SMM Estimation Results: Replication of Cooper & Ejarque (2003) Table 3

Panel A: Structural Parameter Estimates			
Parameter	Our Replication		Original Paper
	Estimate	Std. Err.	Estimate (Table 3)
Curvature (α)	0.7411	(0.0006)	0.696
Adjustment Cost (γ)	0.3518	(0.0019)	0.133
Persistence (ρ)	0.6783	(0.0016)	0.893
Shock Volatility (σ)	0.2478	(0.0007)	0.098
Fixed Fin. Cost ($\tilde{\phi}_0$)	0.0005	(0.0005)	0.000

Panel B: Moments (Data vs. Model)		
Moment	Data Target	Model Fit
Regression coeff on Q (a_1)	0.030	0.125
Regression coeff on Cash Flow (a_2)	0.240	1.162
Serial Correlation of I/K	0.400	0.375
Std. Dev. of Profit Rate (π/K)	0.250	0.091
Mean Average Q ($\bar{q}$)	3.000	2.819
External Finance Fraction	0.250	0.079
J -Statistic	31140.44	

The estimation results yields $\hat{\phi}_0 = 0.0005$ with a standard error of 0.0005. This is statistically indistinguishable from zero and very close to the paper's estimate of 0. This confirms the paper's main conclusion, which was that once market power is accounted for, costly external finance is not required to match investment moments.

1. $\hat{\alpha} \approx 0.74$, which is close to the paper's estimate of 0.696. This confirms the presence of diminishing returns to scale (market power), which is the primary driver of the gap between average Q and marginal Q .
2. While replicating, it ended up with a slightly different structural combination than the authors. The simulation code result shows that higher adjustment costs ($\hat{\gamma} \approx 0.35$ vs 0.13) and lower persistence ($\hat{\rho} \approx 0.68$ vs 0.89). It relied more heavily on adjustment costs (γ) to match the serial correlation target (0.375 vs. target 0.40), whereas the authors relied more on shock persistence (this part should be improved to replicate the results similarly further).

3.1 Numerical Details

The dynamic programming problem is solved using Value Function Iteration (VFI) on a discretised state space.

- The continuous productivity process $\ln A_t$ is approximated by a Markov chain with $N_A = 7$ states using the method of [2]. The capital stock K_t is discretized into a grid of $N_K = 60$ points linearly spaced between $K_{\min} = 1.0$ and $K_{\max} = 350.0$.
- The VFI algorithm terminates when the maximum absolute difference between successive value functions falls below a tolerance of 1×10^{-4} .
- For this experiment, I simulated a panel of $N = 500$ firms for $T = 30$ periods. Continuous uniform random draws are mapped to the discrete productivity states using inverse transform sampling on the Tauchen transition matrix.
- Nelder-Mead simplex algorithm has been employed for the minimisation because the discretisation of the state space via grids leads to a non-smooth SMM objective function.

References

- [1] Russell Cooper and Joao Ejarque. “Financial Frictions and Investment: Requiem in Q”. In: *Review of Economic Dynamics* 6.4 (2003), pp. 710–728. DOI: 10.1016/S1094-2025(03)00047-2.
- [2] George Tauchen. “Finite State Markov-Chain Approximations to Univariate and Vector Autoregressions”. In: *Economics Letters* 20.2 (1986), pp. 177–181. DOI: 10.1016/0165-1765(86)90168-0.